Title: Q-Mutate: A Computational Analysis of Quantum Proton Tunneling in DNA
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This paper presents a quantum mechanical simulation of DNA base-pair stability. By solving the Time-Independent Schrödinger Equation, I calculated the probability of proton transfer in hydrogen bonds, identifying a significant quantum-mechanical pathway for spontaneous genetic mutations at biological temperatures (310 K).

I utilized the Finite Difference Method to discretize the Hamiltonian operator on a spatial grid of 2000 points. The potential energy was modeled as a Quartic Double-Well, representing the chemical environment of a Guanine-Cytosine (G-C) base pair.

The system is governed by the equation:

$\hat{H}\psi = E\psi$Where $\hat{H}$ is the Hamiltonian matrix containing kinetic and potential energy terms. I employed a Sparse Matrix Eigenvalue Solver (scipy.sparse.linalg.eigsh) to determine the ground state and first excited state energies.

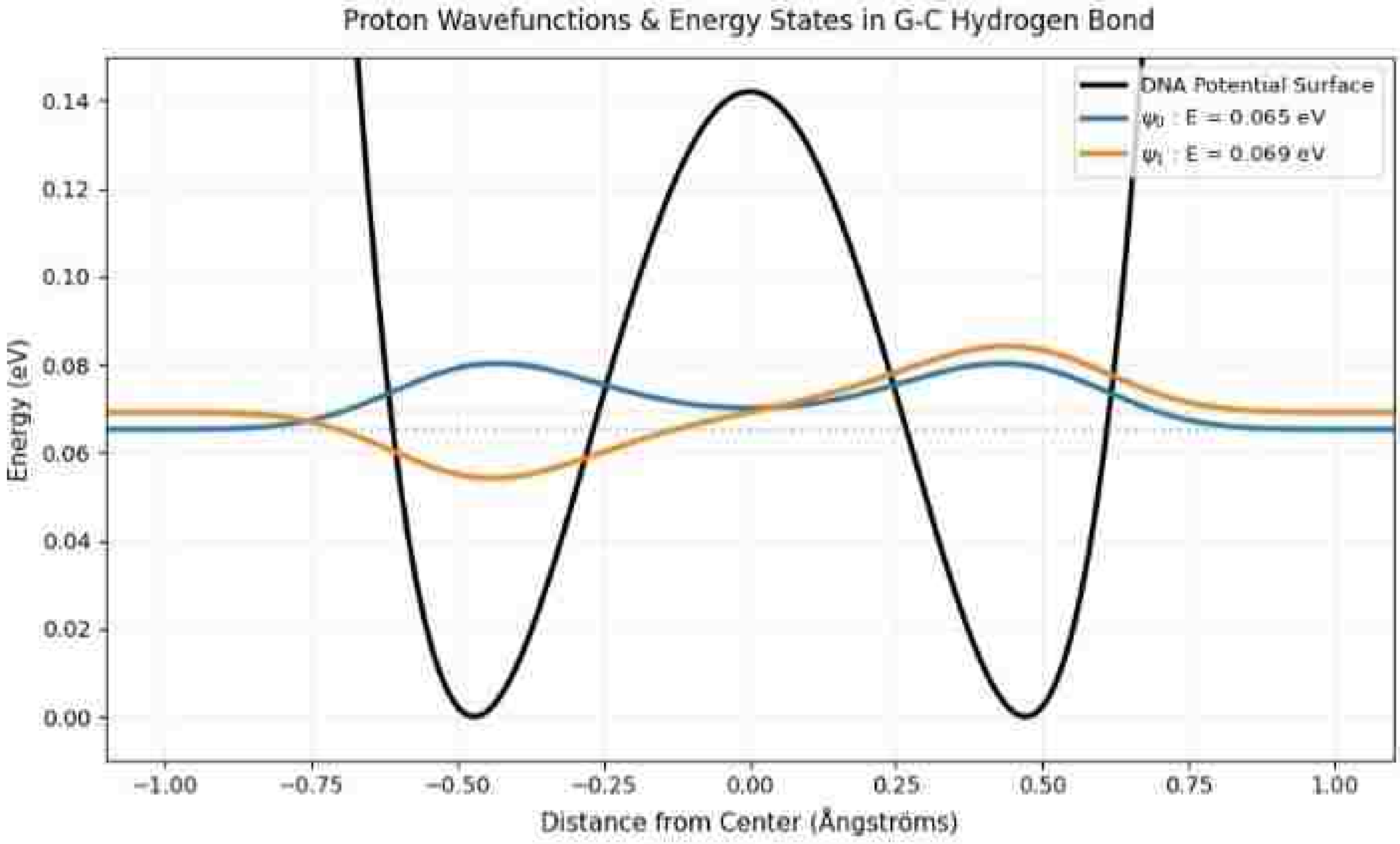
This project confirms that Quantum Tunneling is a viable mechanism for spontaneous point mutations. While classical biology treats DNA as a static database, this research highlights its nature as a dynamic quantum system. Understanding these "Quantum Hotspots" is critical for the future of:

Precision Oncology: Identifying where mutations are most likely to occur.

Quantum Bioinformatics: Developing new algorithms to predict genetic decay.

Future Work:

I intend to expand this model to include Environment-Induced Decoherence, analyzing how the surrounding cellular water molecules affect the tunneling rate.



Proton Wavefunctions & Energy States in G-C Hydrogen Bond
DNA Potential Surface
ψ_0 : E = 0.065 eV
ψ_1 : E = 0.069 eV
Energy (eV)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Distance from Center (Ångströms)